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Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

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ABSTRACT

Image reconstruction for automated optical inspection must improve visibility without converting optical variation, sampling phase, or learned sharpening into false motion. This paper presents MOSR, a measurement-oriented same-size microscopy reconstruction method designed around three metrology failure modes: field- and magnification-dependent optical variation, sampling-phase sensitivity, and reconstruction-induced geometry drift. MOSR addresses them through spatial-variation-aware point-spread function (PSF) guidance, anti-aliased shift-stable reconstruction, and an input-anchored geometry-preserving output. A single set of MOSR weights processes 20x and 50x images with magnification-specific calibrated optical information. Raw, fixed Wiener, and MOSR share one threshold-free maximum-gradient and least-squares circle (LSC) measurement backend. Across four sequences and 120 predefined adjacent-frame pairs, MOSR reduced step standard deviation, nominal-step mean absolute error, and median within-grid SD by 17.9%, 18.0%, and 61.0% relative to Raw, and by 7.8%, 10.6%, and 18.2% relative to fixed Wiener. These results demonstrate improved native-grid positional stability across objective changes without redefining the downstream measurement.

Keywords-Automated optical inspection; measurement-oriented reconstruction; cross-magnification; micro-displacement; positional stability.

I. INTRODUCTION

Image-based optical metrology derives dimensional information from optical image responses and edge-related signals [1]. Periodic circular features such as via- and bump-like arrays are commonly localized during automated optical inspection. When displacement is measured from reconstructed micrographs, reconstruction becomes part of the measurement chain: a visually sharper edge can still move by a fraction of a pixel and change the reported center. Positional stability is therefore required in addition to apparent resolution.

Learning-based microscopy restoration can recover structures that are difficult to inspect in raw data [2]. U-Net-type encoder-decoders combine contextual feature extraction with precise localization [3], but commonly used downsampling operations can introduce shift sensitivity [4]. The problem intensifies when one inspection workflow spans 20x overview images and 50x verification images, because object-space sampling, optical response, and pixel phase change with the objective.

Rather than maximizing visual sharpness, MOSR is organized around three metrology failure modes. Spatial-variation-aware PSF guidance addresses optical variation across field positions and magnifications. Anti-aliased, shift-stable reconstruction is designed to reduce sampling-phase sensitivity. An input-anchored geometry-preserving output limits reconstruction-induced boundary and center drift. A single set of model weights processes both magnifications with their corresponding calibrated optical information.

The contribution is therefore not a new visual-quality objective alone. It is a coordinated reconstruction strategy in which optical guidance constrains correction direction, shift-stable processing limits pixel-phase effects, and the input anchor retains the original geometric reference. The method is evaluated only after these controls are connected to one fixed subpixel measurement backend.

II. METHOD

A. Metrology-driven MOSR design

MOSR receives one monochrome frame and returns a same-size frame on the native coordinate grid. First, calibrated optical information from multiple field locations forms spatial-variation-aware guidance, supplying a bounded optical direction rather than replacing the input by strong inverse filtering. Second, anti-aliased feature expansion and translation-consistency learning are used to reduce position changes caused by decoder sampling. Third, polarity-aware bounded boundary correction is combined with an input anchor so that large-scale intensity, low-frequency geometry, and the original center reference remain tied to the measured frame. Figure 1 summarizes the three design groups and the common measurement backend.

One shared set of MOSR weights processes both 20x and 50x inputs; the optical information changes with the calibrated objective, but the learned reconstruction weights and native-grid output convention remain fixed. A fixed Wiener baseline, based on Wiener filtering [5], uses magnification-specific calibrated PSFs with one regularization setting fixed for all frames.

These components are designed to reduce sampling-phase sensitivity and geometry drift, and their effects are assessed through the common boundary-and-center backend.

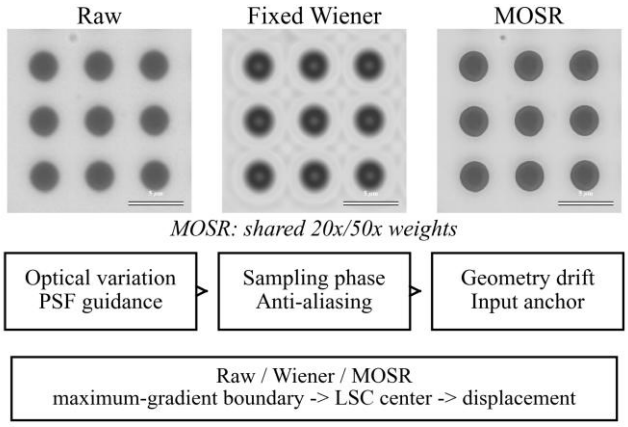


FIGURE 1. REPRESENTATIVE 50X OUTPUTS (TOP; SCALE BAR, 5 MICROMETERS), COMPACT MOSR DESIGN SUMMARY, AND THE COMMON MEASUREMENT BACKEND.

B. Common boundary, center, and metrics

All methods use the same threshold-free measurement backend. A Chan-Vese active contour provides only a closed coarse curve [6].

Profiles are sampled along its outward normals, and the correct-polarity gradient maximum is refined to a subpixel position by local quadratic interpolation [7]. These points, rather than the regularized active-contour pixels, define the measured boundary.

Let (xi, yi) denote the accepted points. A geometric least-squares circle (LSC) [8] supplies the center and radius by minimizing the sum of squared radial residuals; stage commands and crop initialization do not enter the fit. The common estimator was verified with synthetic circles, known translations, and perturbed crop initializations before comparison.

For each frame pair, the median of nine pillar shifts represents the array step, while their standard deviation defines the within-grid SD. Aggregate Step SD and nominal MAE are the means of the four condition-level values; Median within-grid SD is the median of the 120 pair-level nine-pillar standard deviations. The three criteria remain separate rather than being combined into an adjustable score.

III. EXPERIMENT AND RESULTS

A. Locked cross-magnification experiment

Four displacement sequences were recorded from a periodic 3x3 pillar array using 20x and 50x objectives at nominal 0.5 and 1.0 micrometer steps. Their object-space sampling intervals were 0.1725 and 0.0690 micrometers/pixel. The 120 adjacent-frame pairs were predefined using the raw acquisition sequences before any reconstruction results were inspected and were then held fixed for all methods. Raw, Wiener, and MOSR used the same images, nine features, scale, boundary, and circle center.

All images retained the same contrast polarity, and no method-specific edge cleanup was introduced. Pixel coordinates were converted to object space only after the circle centers had been estimated. Reconstruction comparisons are therefore not affected by different interpolation, threshold, or scale-conversion procedures.

B. Measurement performance

MOSR achieved the lowest aggregate value for every reported criterion (Table I and Fig. 2). Relative to Raw, Step SD, nominal MAE, and Median within-grid SD decreased by 17.9%, 18.0%, and 61.0%. Relative to fixed Wiener, the reductions were 7.8%, 10.6%, and 18.2%. MOSR returned the lowest within-grid SD in all four imaging conditions and the lowest Step SD and nominal MAE in three of four conditions. At both magnifications, its Step SD remained near 0.06 native pixel.

At 50x, MOSR reduced the Wiener magnification-level Step SD, nominal MAE, and Median within-grid SD from 4.83, 4.37, and 3.41 nm to 4.51, 4.02, and 2.56 nm. At 20x, the corresponding values decreased from 10.98, 8.47, and 13.11 nm to 10.07, 7.46, and 7.55 nm. The larger 20x nanometer values are consistent with its coarser sampling, while the fractional-pixel Step SD remains comparable across objectives.

In a matched upsampling comparison, replacing anti-aliased upsampling increased Median within-grid SD by 0.27 nm, with a 95% paired resampling interval of 0.05-0.74 nm, while changes in Step SD and nominal MAE remained below 0.1 nm. This localizes the supported role of anti-aliasing to cross-position consistency rather than attributing every MOSR improvement to one component.

Across the full method, all six 95% paired baseline-minus-MOSR intervals remained above zero. MOSR produced lower within-grid SD than Raw in 119 of 120 comparisons and lower within-grid SD than Wiener in 92 of 120 comparisons. The conclusions concern comparative in-system stability rather than external length traceability or cross-device generalization.

TABLE I. AGGREGATE DISPLACEMENT RESULTS ACROSS 120 PAIRS (NM).

Method	Step SD	Nom. MAE	Median within-grid SD
Raw	8.88	7.00	12.55
Wiener	7.91	6.42	5.98
MOSR	**7.29**	**5.74**	**4.89**

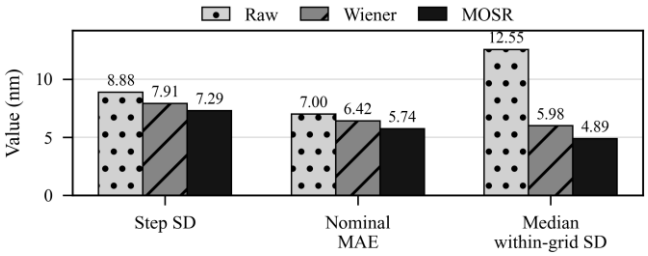


FIGURE 2. AGGREGATE DISPLACEMENT METRICS ACROSS 120 PAIRS; LOWER VALUES INDICATE BETTER PERFORMANCE.

IV. CONCLUSION

MOSR combines spatial-variation-aware PSF guidance, anti-aliased shift-stable reconstruction, and an input-anchored geometry-preserving output. With one set of weights and one common measurement backend, the method achieved the best aggregate repeatability, nominal agreement, and within-grid consistency across four 20x/50x sequences and 120 predefined pairs. MOSR therefore improves reconstruction for cross-magnification subpixel position measurement rather than pursuing visual sharpness alone.

The shared native-grid convention also preserves a common position definition when the inspection objective changes, supporting repeatable alignment and registration of periodic circular features in the tested system.

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